

Survey Design Seminar Survey Methods 670

Winter 2026

Week 5: Alternative Research Designs

February 5, 2026

Outline of Today's Class

The Scientific Method

Reliability and Validity in Measurement

Steps in the Research Process

Different Research Designs

Basic Principles Embodied in the Scientific Method and Empirical Research

- 1. Based upon observation**
- 2. Methods are objective and unbiased**
- 3. Knowledge sought is general**
- 4. Process is logical**
- 5. Hypotheses are falsifiable**
- 7. Goals are explanation and prediction**
- 8. Simple (parsimonious) explanations are preferred**

How Do These Considerations Affect Consultants?

Time may not be on your side



How does the client's calendar fit into your research plan?

Secondary analysis

Meta-analysis

Steps in the Research Process

Conceptualization



Operationalization



Data Collection



Data Analysis



Conclusions

Original Data Collection Vs. Secondary Data Analysis

Why would we use someone else's data?

- **Maybe there is no chance of independent data collection (Budget? Time?)**
- **Could use better data (sampling, measures) than we could collect on our own**
- **Potential ease of access, rapid turnaround**

Tradeoffs in Original Data Collection vs. Secondary Data Analysis

ORIGINAL DATA COLLECTION

Lengthy field period

Sample design and selection

Operationalize own concepts

Write own questions/
responses

Employ desired levels of
measurement

SECONDARY DATA ANALYSIS

Immediate data analysis

Deal with available sample

Use available concepts

Use available measures

Employ available levels of
measurement

The Difference Between a Research Question and an Operational Hypothesis

Q: What is the effect of interviewing mode on data quality?

H: Interviews conducted with live interviewers will have a higher data quality than those which are not.

Selecting a Dataset: Choosing among Available Alternatives

Design of the study

- Select research design(s)

- Collect data from a sample / population

Specific measures in the questionnaire

Access and availability

Meta-analysis

An approach that involves the **analysis of analyses**

Alternative approaches:

Identification of independent and dependent variables

Vote counting procedure:

$$\text{Success Rate} = (\# \text{ Successes} / \# \text{ Tests}) * 100$$

$$\text{Effect Size Proxy} = \frac{(\# \text{ Successes} - \# \text{ Anomalies})}{\# \text{ Tests}}$$

for each IV

There Are Three Criteria for Establishing a Causal Relationship

1. Covariation

- When the values on one variable change, so do values on the other
- Variables **CHANGE TOGETHER**, they are **CO-RELATED**

2. Temporal Order

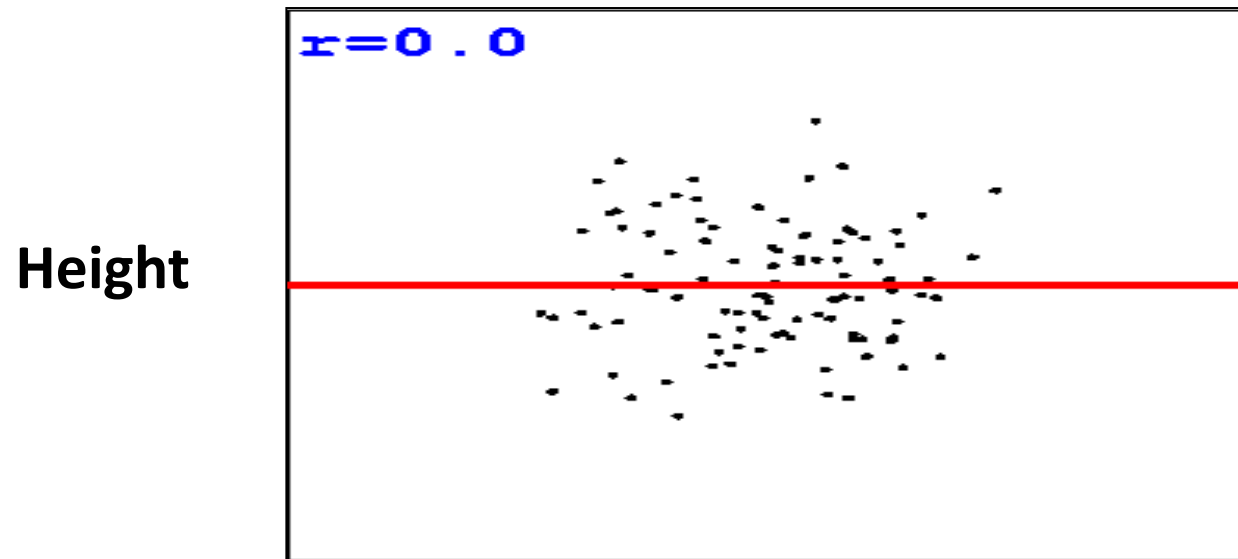
- Changes on the independent variable precede changes on the dependent variable in time. X is the cause of Y and not the reverse.

3. Elimination of rival hypotheses

- Other potential causes are accounted for and ruled out. **PLAUSIBLE ALTERNATIVE HYPOTHESES** are eliminated

The Nature Of Relationships

Magnitude: How STRONG is the relationship?



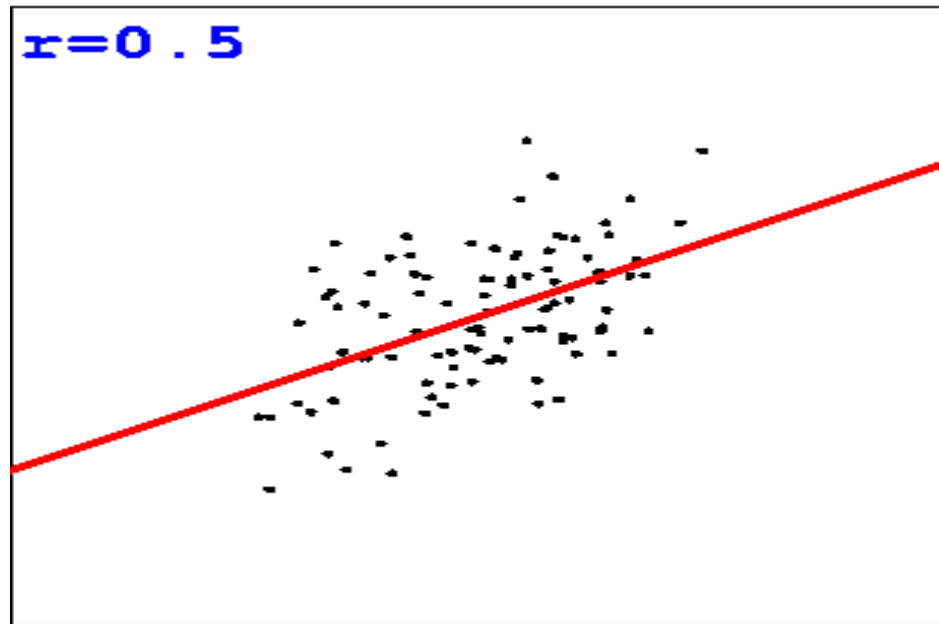
Television Exposure

Correlation (r) = .0

The Nature Of Relationships

Magnitude: How STRONG is the relationship?

**Attention to
Campaign
News**



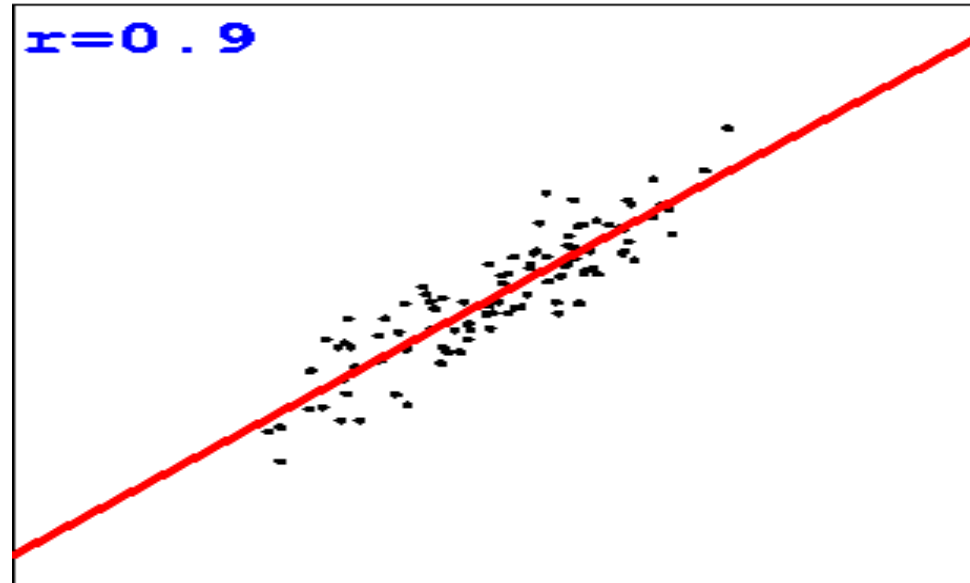
Political Participation

Correlation (r) = .5

The Nature Of Relationships

Magnitude: How STRONG is the relationship?

**Ads
Broadcast**



Campaign Contributions

Correlation (r) = .95

Covariation Is Necessary but NOT SUFFICIENT for Causality to Occur

- **When my knee hurts, it rains.**

When my knee hurts, it rains.

Knee Condition



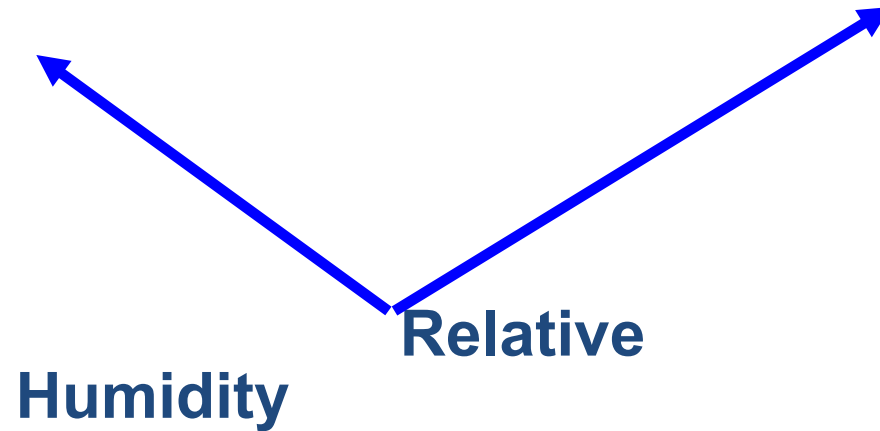
Rain

Spurious Relationship

When my knee hurts, it rains.

Knee condition

Rain



Other problems with causal claims:

H: Listening to Political Talk Radio Causes You to Become Conservative.

- Evidence: People who listen to Steve Bannon are more conservative than people who do not
- Alternative hypothesis?

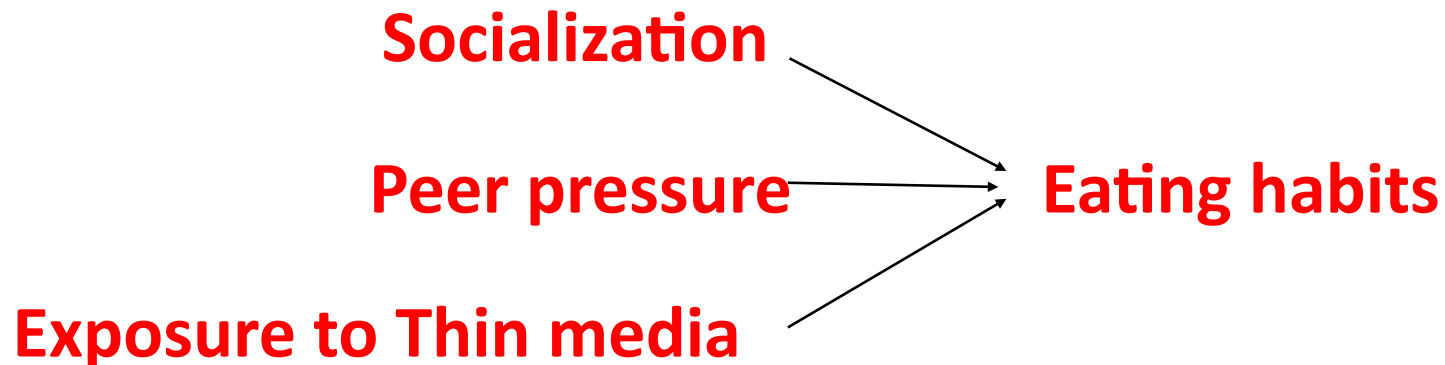


Different Ways of Looking at Causation

Monocausal explanations

- HIV and AIDS

Multi-causal explanations



How to Examine Assertions or Claims

- 1. State the question/problem simply: Is there a hypothesis?**
- 2. Identify the key concepts**
- 3. Identify the hypotheses that link the concepts together**
- 4. Look at the explanations**
 - *Are causal relationships stated or implied?**
 - *What kind of causal relationships make sense?**
 - *Are any important variables left out, i.e., intervening, conditional, or explanatory?**
- 5. Are the necessary conditions for causality satisfied?**
 - Covariation**
 - Temporal order**
 - Elimination of Rival or Plausible Alternative Hypotheses (ERH or PAH)**

Types of Errors

The scientific method is organized around minimizing the chance of making certain types of errors.

Sources of Error in Observation: Measurement Errors

Random errors are due to chance fluctuations and average to zero

Systematic errors are not due to chance and they have a direction or "*bias*"

Where Does Random Error Come From?

Derives from 2 sources:

- 1. Natural random variations**
- 2. Imprecise measurement/ inadequate instruments**

Where Does Systematic Error Come From?

Could derive from a number of sources:

- 1. Researchers**
- 2. Subjects**
- 3. Researcher-subject interactions**
- 4. Subject selection/ sampling**

How Researchers Deal with Random Measurement Error

- 1. Careful conceptualization**
- 2. Careful use of statistical inference**
- 3. Precise measurement**
- 4. Use of multiple measures for the concept**

Searching for random error and assuring ourselves that it is minimized involves the notion of RELIABILITY

How Researchers Deal with Systematic Measurement Error

- 1. Researchers (Assistants) are blind to the Hypothesis**
- 2. Subjects are blind to the Hypothesis**
- 3. Both researchers and subjects are blind to the Hypothesis (double blind)**
- 4. Careful design of measures and checking against other known measures of the same concept**

Searching for systematic error and assuring ourselves that it is minimized involves the notion of VALIDITY

RELIABILITY is concerned with precision and consistency in measurement.

It can involve a relationship between repeated measurements of the same concept, in the form of a hypothesis:

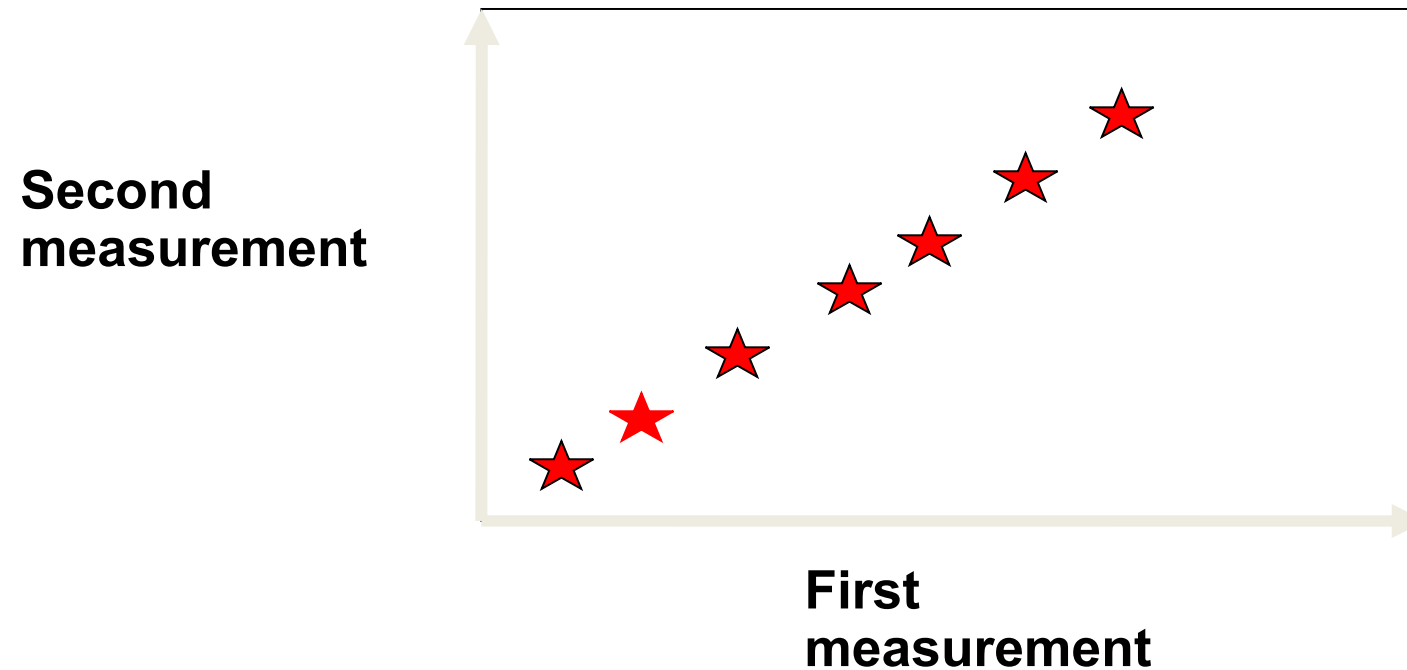
Measurement Theory

H: Repeated measures of the same concept using the same operationalization should return the same value, on average.

*** Over time**

*** Over measuring devices
(instruments or people)**

Expected Observations with Repeated Measurement



Standard Assumption about Random Errors

The only difference between two measure(ment)s of the same concept should be random

Random errors:

- 1. Are both positive AND negative**
- 2. Sum to zero**

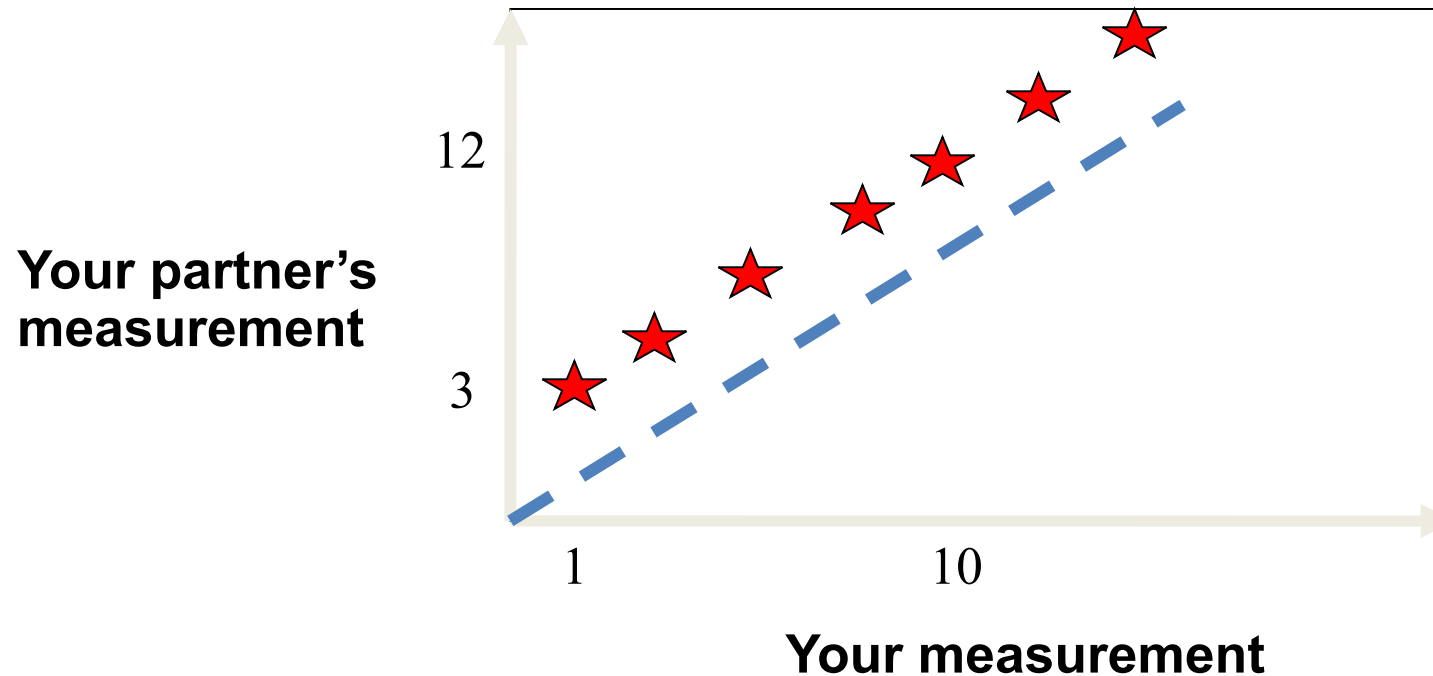
Standard Assumption about Systematic Errors (Bias)

1. They tend to be either positive OR negative.
2. They DO NOT sum to 0.

You and another person are going to code video segments on a “violence” scale

You and a partner watch the same segments, but he is more sensitive to violence than you are.

Repeated Measurement with Systematic Error (Bias)



What does this say about your and your partner's stop watches? Or your measuring abilities?

Choosing a Research Design

The RESEARCH DESIGN consists of the specification of the details of data collection – how the units of analysis will be assembled and how the units of measurement will be recorded.

It is the method for collecting the data or making the observations.

The RESEARCH DESIGN deals with three fundamental issues of explanation and prediction (the goals of science):

- 1. Making valid and reliable measurements**
- 2. Establishing causation**
- 3. Generalizing to the real world**

Evaluating Alternative Observation Methods

The ways we collect data have implications for all three of these issues.

- 1. Direct vs. indirect**
- 2. Structured vs. unstructured**
- 3. Obtrusive vs. unobtrusive**
- 4. Participatory vs. non-participatory**
- 5. Manipulative vs. non-manipulative**

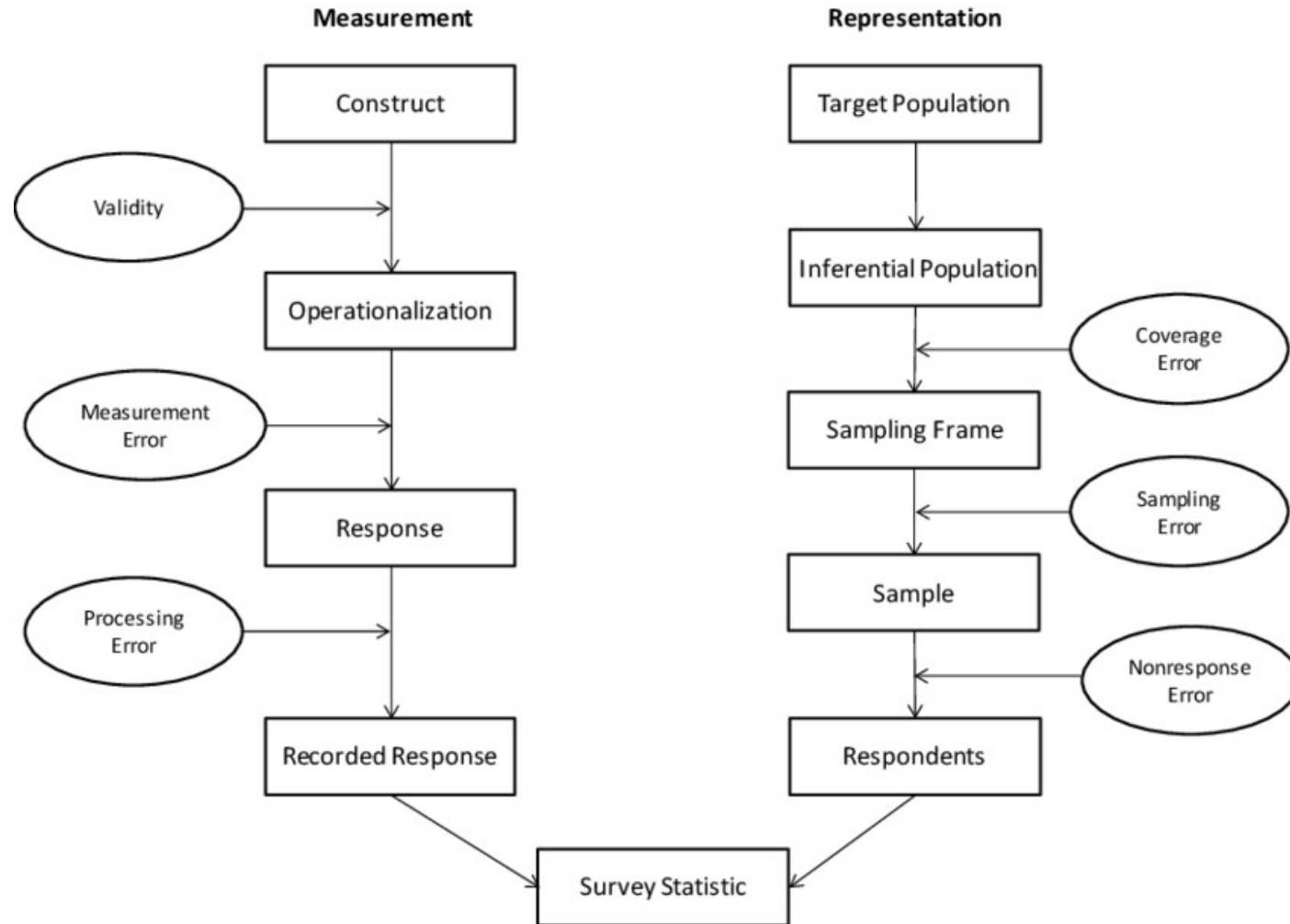
All research involves a series of cost-benefit calculations because no researcher has an infinite (unlimited) amount of resources (time or money), or even as much as they would like

For different projects, we could be willing to accommodate different kinds or levels of error

So researchers have to make decisions such as for how many variables and how many “cases” (units of analysis or subject units) they should collect data – OLD VERSION

And they also have to decide how much to invest in “collecting data” (making the observations) as opposed to how much to invest in “error reduction” (reliability and validity test) – NEW VERSION TSE

The Total Survey Error (TSE) Model



Design of Experiments

The experimental design begins in the laboratory under controlled conditions (physical science):

This means that the researcher

- 1) Establishes the conditions that are the equivalent of the independent variable**
- 2) Observes the value of the dependent variable before and after the independent variable is applied**
- 3) Knows that the independent variable precedes the change in the value of the dependent variable**

Choosing a Research Design

We pick a design once we have described the phenomenon and developed the conceptual model

Remember the goals:

- 1. Reliable and valid measures**
- 2. Establish cause**
- 3. Generalize to the real world**

How Do We Evaluate the Quality of Our Design?

DESIGN VALIDITY comes in two forms:
Internal and External.

Internal Validity

- 1. Reliable and valid measures**
- 2. Establish causality:**
Covariation, temporal order,
Elimination of Rival Hypotheses

External Validity

- 3. Generalize from our study to
the real world**

Note: Design validity *incorporates* measurement validity

Internal Validity

- 1. The conclusions drawn from the analysis accurately reflect what went on during the period of observation**
- 2. The conditions for establishing causality were fulfilled**

Threats: Alternative hypotheses/ other possible independent variables, poor measurement, can't determine temporal order, improper data analysis

Threats to Internal Validity

- **Mortality:** Did certain types of subjects drop out of the study?
- **History:** Other events between T1 and T2?
- **Maturation:** Did the subjects get older and change over time?
- **Instrumentation:** Changes in measurement
- **Regression to the mean:** Extreme scores are likely to be less extreme when measured again
- **Testing:** Could the act of the pretest affect the post-test measurement (sensitization)?

External Validity

The conclusions drawn from the analysis apply in the “real world”

Threats:

- 1. The subjects are different than the population you are trying to understand**
- 2. The setting for the data collection is not “naturalistic”**

Threats to External Validity

- 1. The units of analysis are different than the ones to which the researcher is trying to generalize in the “real world.”**

Are college sophomores the same as “real people”?

NOTE: Only a problem if they differ in ways that would affect the relationship!

2. The setting in which the observations were made is not like the “typical” setting in which the relationship might arise in the “real world”

Watching television in a group with strangers in a lab setting versus watching at home

Again: Would the setting affect the relationship being studied?

Evaluating Research Designs in Others' Work

1. Internal validity

Does the study employ valid and reliable measures?

Does the study satisfy the conditions for establishing causality?

2. External validity

Did they study the correct units?

Does the setting correspond to the real world?

Evaluating Research Designs

- 1. How well do they deal with internal and external validity?**
- 2. How do they employ various types and kinds of measurement?**
- 3. What kinds of units of analysis are typically used?**
- 4. What do they cost?**

Reinterpreting the Hypothesis

Saying that there is an expected relationship between X and Y means that the value of Y changes when different values of X occur (a difference in the value of Y after X - temporal order)

It also implies that there is no change in Y if X does not occur (NULL HYPOTHESIS)

Example: What is the impact of mode on data quality?



How Could This Study Be Organized as an Experiment?

Randomization:

Manipulation:

Control:

What Are the Strengths and Weaknesses of Such a Design?

This is a question that we answer by talking about:

Internal validity

External validity

What are the issues typically evaluated here?

The Elements of the Quasi-experimental Design

EXPERIMENT

- 1. Randomization**
- 2. Manipulation**
- 3. Control**

QUASI- EXPERIMENT

- 1. Matching**
- 2. Manipulation**
- 3. Control**

What Does This Mean?

Matching

Finding two settings that are similar on as many dimensions as possible, implying the units of analysis will be similar

Manipulation

Applying a treatment in a way that permits pre and post-measurement of the dependent variable

Control

Having a sample in which there is no “treatment”

How to Approach This Problem in the Data Quality Context

Collect data from surveys where the mode has changed (from phone to web or landline to cell)

Look at data quality (how to measure?) as a result

Need to look at more than just 2 time points (pre and post) to make sure effects are durable

What is the list of “plausible alternative explanations” over the last decade?

Think of Plausible Alternative Explanations for Changes in Data Quality

Response rates

Resulting sample composition

Instrumentation changes

???????

There are two kinds of plausible alternative hypotheses:

1. Those that could exist on some theoretical or conceptual basis, i.e. be reasonably known in advance.

These can be dealt with by including another variable in the study (PAH)

2. Those that are unknown but might exist

These are dealt with in experiments through **RANDOMIZATION**, i.e., making all other things equal

An experiment including appropriate randomization deals effectively with PAH's, especially the "unknown" kind